

Two Samples, One Difference

Unit 4 · Topic 4.6 · TODAY'S PROBLEM

Suppose parcel pickup waits have the shown population parameters, measured in minutes. There are 600 North visits and 1,000 South visits. North waits are approximately normal; South waits are strongly right-skewed. Independent SRSs are taken without replacement, and the service wants $P(\bar{X}_N - \bar{X}_S > 2.5)$.

Rina: “South is skewed, so I would not use a normal model for the difference.”

Oscar: “North is normal and the combined sample size is 60, so the normal model is valid; the probability is 0.0273.”

Locker samples

Locker	n	μ	σ
North	12	6.0	1.8
South	48	4.8	3.0

FIRST TAKE (5 min, on your sheet)

Would you check each place's sample size separately or add them together? Give one reason from the study.

- DOK 1** (a) State the mean of $\bar{X}_N - \bar{X}_S$ and translate the event above into context.
- DOK 2** (b) Verify both 10 percent conditions, compute the sampling-distribution SD, and calculate the probability under a normal model.
- DOK 3** (c) In 4–5 sentences, decide whether 0.0273 is supported and explain what each student overlooks. Justify each sample mean using its own shape and size. Then consider $n_N = 48$ and $n_S = 8$: explain why their total alone cannot justify the same normal-model claim.

Optional review: [follow-along](#) (scan the code)

Work (a)–(c) from the rules on your sheet. Turn in your sheet.

